

MBE-Net: A Multi-Branch Edge-Guided Feature Enhancement Algorithm for SAR Ship Detection

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Abstract—Synthetic Aperture Radar (SAR) imagery plays a vital role in maritime surveillance, yet it faces severe challenges such as speckle noise, cluttered backgrounds, and the small scale and low contrast of ship targets. To overcome these issues, we propose MBE-Net, a lightweight and high-accuracy SAR ship detection network featuring multi-branch edge-guided feature enhancement. Firstly, the Edge-Enhanced Wavelet Backbone (EEWB) is designed to enhance edge information and suppress detail loss by integrating multi-branch edge guidance and wavelet pooling, thereby improving small-ship perception in complex scenes. Secondly, the Lightweight Decoupled Detail-Enhanced Head (LD-DEH) decouples classification and regression tasks while embedding a detail-enhanced convolution module to recover high-resolution features, significantly boosting localization precision under strong interference. Lastly, the Adaptive Contextual Focus Feature Pyramid Network (ACF-FPN) introduces context-guided fusion to improve multi-scale feature alignment and robustness, especially in low-contrast and cluttered environments. Extensive experiments on the HRSID and SSDD datasets validate that MBE-Net achieves excellent detection accuracy with minimal computational cost, making it highly suitable for real-time SAR ship detection.

Index Terms—SAR ship detection, lightweight detection network, wavelet pooling, multi-scale feature fusion, edge enhancement.

I. INTRODUCTION

SYNTHETIC Aperture Radar (SAR) is a powerful remote sensing technology that provides high-resolution images of the Earth's surface, particularly useful for ship detection in marine environments. SAR systems can operate in all weather conditions and during both day and night, offering significant advantages over optical imaging methods. However, detecting ship targets in SAR images remains challenging due to factors such as small target sizes, complex backgrounds, and speckle noise [1].

Traditional ship detection methods, including handcrafted

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features and machine learning models, often suffer from high false positive and negative rates, particularly in complex backgrounds. Recently, deep learning-based approaches, such as Faster R-CNN, SSD, and YOLO, have shown promise for SAR ship detection. Nevertheless, these approaches still suffer from intense computational demand, small target detection challenges, and degraded accuracy in dense or cluttered scenes. While YOLO algorithms balance speed and accuracy, their ability to handle complex backgrounds and small targets still requires improvement [2].

Recent improvements include SMEP-DETR, proposed by Chushi Yu et al. [3], which integrates speckle denoising, multi-edge enhancement, and a parallel dilated convolution–attention pyramid network into a transformer framework, greatly improving robustness in complex SAR scenes. Hang Yu et al. [4] developed LD-Det, a lightweight YOLOv8-based detector that uses dual-domain feature fusion, partial convolution, and a triple attention mechanism to balance accuracy and efficiency. Yongbo Ma et al. [5] proposed 3SD-Net, combining bidirectional coordinate attention (BCA), multi-resolution feature fusion (MRF), and a center point distribution module (CPDM) to enhance detection, especially for small ships with varying aspect ratios. Despite these advancements, challenges such as high computational complexity, insufficient small target detection capabilities, and difficulties in detecting SAR ship targets in complex marine backgrounds remain.

To address these challenges, this paper proposes MBE-Net (Multi-Branch Edge-Guided Network), a lightweight and high-accuracy SAR ship detection algorithm. Its core components include: an Edge-Enhanced Wavelet Backbone (EEWB) for strengthening edge representation and preserving high-frequency details; a Lightweight Decoupled Detail-Enhanced Head (LD-DEH) that incorporates a three-stage enhancement module and supports inference-time weight fusion to efficiently recover spatial resolution; and an Adaptive Contextual Focus Feature Pyramid Network (ACF-FPN) for enhancing multi-scale feature fusion, edge preservation, and robustness in cluttered environments. Experimental results demonstrate that MBE-Net delivers competitive accuracy in detecting small ships under complex marine conditions.

II. PROPOSED METHOD

This section provides a detailed overview of MBE-Net, including the Edge-Enhanced Wavelet Backbone (EEWB), the Lightweight Decoupled Detail-Enhanced Head (LD-DEH), and the Adaptive Contextual Focus Feature Pyramid Network (ACF-FPN). The overall architecture is illustrated in Figure 1.

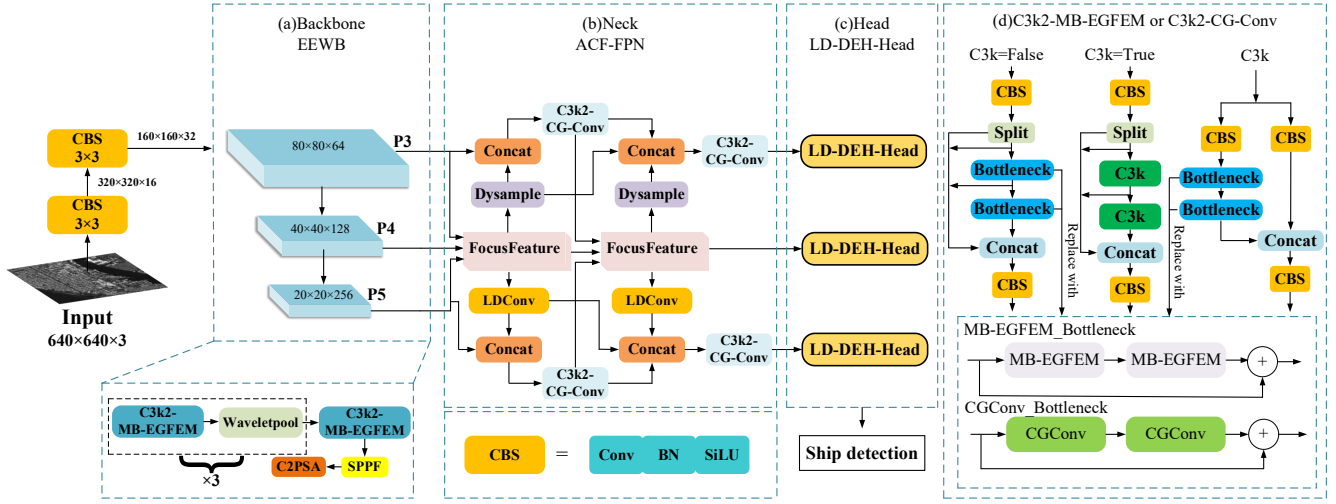


Fig. 1. Overall architecture of MBE-Net based on YOLO11n, showing (a) the proposed Edge-Enhanced Wavelet Backbone (EEWB), (b) the Adaptive Contextual Focus Feature Pyramid Network (ACF-FPN), and (c) the Lightweight Decoupled Detail-Enhanced Head (LD-DEH). (d) The structure of the baseline C3k2 bottleneck and its replacements, where the original bottleneck is replaced by MB-EGFEM_Bottleneck or CGConv_Bottleneck to form C3k2-MB-EGFEM or C3k2-CG-Conv.

A. Edge-Enhanced Wavelet Backbone (EEWB)

In SAR ship detection, blurred contours, weak edges, and detail loss caused by downsampling severely hinder small-ship detection. To address these issues, we design the Edge-Enhanced Wavelet Backbone (EEWB) in MBE-Net, which integrates the Multi-Branch Edge-Guided Feature Enhancement Module (MB-EGFEM) and WaveletPool [6].

MB-EGFEM enhances edges and contours via spatial-domain multi-scale fusion and edge-guided refinement, while WaveletPool preserves fine details by frequency-domain decomposition using fixed Haar filters. The two modules are complementary: MB-EGFEM emphasizes structural boundaries, while the high-frequency details retained by WaveletPool reinforce contour extraction, thereby improving small-ship detection in cluttered SAR backgrounds.

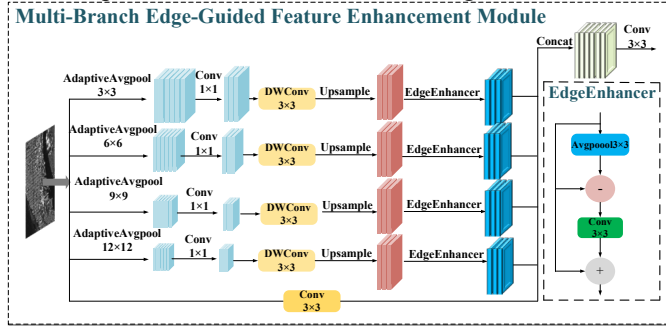


Fig. 2. Architecture of the Multi-Branch Edge-Guided Feature Enhancement Module.

The proposed MB-EGFEM consists of a local feature branch, multi-scale contextual branches, and edge-guided enhancement, followed by a final 3×3 fusion convolution.

Given the input feature F_{in} , the local branch preserves fine-grained spatial details:

$$F_{local} = \text{Conv}_{3 \times 3}(F_{in}) \quad (1)$$

The multi-scale context branch employs four adaptive average pooling scales $S = \{3, 6, 9, 12\}$. For each scale s , pooled features are compressed by a 1×1 convolution,

processed with a depthwise 3×3 convolution, and bilinearly upsampled to the original size:

$$\hat{F}_s = \text{Up}(\text{DWConv}_{3 \times 3}(\text{Conv}_{1 \times 1}(\text{AAPool}_s(F_{in})))) \quad (2)$$

Each branch output is refined by EdgeEnhancer, which extracts low-frequency components via 3×3 average pooling, computes the high-frequency residual, and applies a sigmoid-gated convolution:

$$E_s = \hat{F}_s - \text{Pool}_{3 \times 3}(\hat{F}_s) \quad (3)$$

$$\hat{F}_s^{ee} = \hat{F}_s + \sigma(\text{Conv}(E_s)) \quad (4)$$

Finally, the local feature and all enhanced multi-scale features are concatenated along the channel dimension and fused with a 3×3 convolution:

$$F_{out} = \text{Conv}_{3 \times 3}(\text{Concat}(F_{local}, \{\hat{F}_s^{ee}\}_{s \in S})) \quad (5)$$

This design integrates multi-scale contextual modeling with edge-guided enhancement to preserve fine details and improve ship contour perception in complex SAR backgrounds, as illustrated in Fig. 2.

WaveletPool performs frequency-domain feature downsampling via the Discrete Wavelet Transform (DWT), decomposing features into low-frequency structural components and high-frequency details. This decomposition preserves fine-grained edges while reducing aliasing, making it particularly suitable for detecting small targets in cluttered SAR backgrounds where ships often exhibit weak edges and low contrast. Given an input feature map X , four fixed Haar wavelet filters (LL, LH, HL, and HH) are applied to capture low- and high-frequency information along different orientations:

$$\text{LL} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \text{LH} = \begin{bmatrix} -0.5 & -0.5 \\ 0.5 & 0.5 \end{bmatrix} \text{HL} = \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix} \text{HH} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} \quad (6)$$

The four directional components are computed via grouped 2D convolution and downsampled with stride 2:

$$Y = \text{Conv2D}(X, W, \text{stride} = 2) \quad (7)$$

where W represents the set of wavelet filters expanded across all channels. Grouped convolution ensures that each channel undergoes an independent transformation, reducing

computational cost while enabling multi-scale directional feature extraction.

Within EEWB, WaveletPool complements MB-EGFEM by retaining high-frequency details and providing rich directional cues. These preserved details reinforce MB-EGFEM's spatial edge enhancement, enabling the network to capture clearer ship contours and improve small-ship detection accuracy and robustness in complex marine environments.

B. Lightweight Decoupled Detail-Enhanced Head (LD-DEH)

SAR ship detection is significantly affected by sea clutter, speckle noise, and fine-structure loss caused by repeated downsampling, resulting in blurred boundaries and cross-scale feature drift. To address these challenges, we propose the Lightweight Decoupled Detail-Enhanced Head (LD-DEH), a fully decoupled architecture that separates classification and regression to reduce inter-task interference. At each feature pyramid scale, features are first mapped to a unified hidden dimension via a 1×1 Conv_GN and then processed by a three-stage DEConv_GN stack with shared parameters across scales to enhance edges, contours, and structural details. The refined features are finally fed into independent regression and classification branches, improving detection robustness while maintaining computational efficiency.

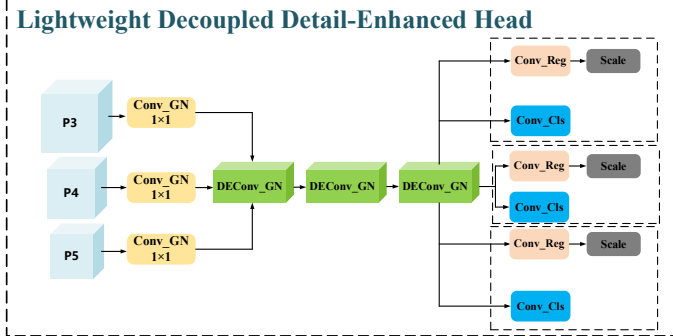


Fig. 3. Architecture of the Lightweight Decoupled Detail-Enhanced Head.

Let the input feature at scale i be $\mathbf{F}_i \in \mathbb{R}^{C \times H_i \times W_i}$. The shared detail-enhanced convolution is formulated as

$$\mathbf{F}'_i = \phi_{\text{DEConv}_G}(F_i; \theta_s), \quad i = 1, \dots, N \quad (8)$$

where θ_s denotes parameters shared across all scales. Each DEConv_GN replaces Batch Normalization with Group Normalization (16 groups) and aggregates complementary kernels—center-difference, horizontal-vertical difference, adaptive difference, and a standard 3×3 convolution—prior to activation:

$$\phi_{\text{DEConv}_G}(F) = \sigma \left(\text{GN} \left(\sum_{k=1}^4 \text{Conv}_k(F) + \text{Conv}_{\text{std}}(F) \right) \right) \quad (9)$$

This multi-kernel design enhances directional gradients and center-surround contrasts, effectively sharpening weak edges and blurred contours that are typical in SAR small-ship targets. During inference, the multiple difference-based convolution branches are re-parameterized into a single equivalent 3×3 convolution by linearly summing their kernel weights and biases, resulting in a standard convolution operation with no additional computational overhead while preserving the learned detail-enhancement behavior.

In the regression branch, a lightweight scale adaptation is applied per level:

$$B_i = \alpha_i \cdot R_i \quad (10)$$

where R_i is the raw DFL regression output (decoded later) and α_i is a learnable scalar mitigating resolution-dependent bias, thereby improving localization consistency for small targets across pyramid levels.

Fig. 3 illustrates its structure. This design improves inference efficiency by re-parameterizing multi-branch convolutions into a single equivalent kernel, reducing computational cost while preserving detail-enhancement capability. Benefiting from cross-scale parameter sharing and decoupled regression, LD-DEH achieves a favorable balance between efficiency and accuracy. As shown in Table I, LD-DEH improves FPS by 22.4 over the YOLO11n baseline. The FPS is measured using Ultralytics validation profiling on a single NVIDIA GeForce RTX 2060 GPU with 640×640 input resolution and a batch size of 16, based on average per-image inference time.

TABLE I
COMPARATIVE EXPERIMENT ON FPS BETWEEN BASELINE AND BASELINE+LD-DEH.

Models	FPS
Baseline	140.3
Baseline+LD-DEH	162.7

C. Adaptive Contextual Focus Feature Pyramid Network (ACF-FPN)

Traditional Feature Pyramid Networks (FPN) are widely used for multi-scale feature fusion in object detection but have limitations in SAR ship detection, especially with cluttered backgrounds, low contrast, and boundary blurring caused by downsampling. The fixed top-down fusion in FPN struggles to adaptively highlight informative features or suppress background interference, limiting its effectiveness for small or irregularly shaped targets.

To address these challenges, we propose the Adaptive Contextual Focus Feature Pyramid Network (ACF-FPN), an enhanced variant of FPN tailored for SAR ship detection. While the core modules of ACF-FPN—FocusFeature, LDConv [7], and DySample [8]—are adapted from existing designs, their strategic combination and optimization significantly improve performance. FocusFeature strengthens edge and local detail representation, essential for ships with blurred contours, while LDConv adapts convolutional receptive fields to handle irregular shapes. DySample restores high-resolution details, preserving small target information, and ContextGuided Convolution [9] refines feature fusion by utilizing both local and global contextual information, enhancing target localization in cluttered backgrounds.

These components are integrated into a shared adaptive fusion framework that selectively emphasizes relevant features while maintaining computational efficiency. Combined with LD-DEH and EEWB in MBE-Net, ACF-FPN facilitates fine-detail enhancement, improved feature discrimination, and effective noise suppression, thereby improving detection accuracy in SAR imagery with complex sea clutter. The synergy of adaptive fusion, deformable convolutions, and context-aware feature fusion enables ACF-FPN to adapt to varying target scales and backgrounds, resulting in robust and accurate SAR ship detection.

III. EXPERIMENT

Experiments are conducted on a Windows 10 workstation with an NVIDIA GeForce RTX 2060 GPU (6 GB) and an AMD Ryzen 7 4800H CPU, using PyTorch 1.12.0 with CUDA 11.3 and Python 3.9.

A. Dataset and Experimental Settings

HRSID [10] contains 5,604 SAR images with 16,951 ships, and SSDD [11] contains 1,160 images with 2,456 ships. Both datasets are randomly divided at the image level into training, validation, and test sets with a ratio of 70%, 10%, and 20%, respectively, with no overlap among subsets.

All models are trained for 350 epochs using SGD with an initial learning rate of 0.01, momentum of 0.937, and weight

decay of 0.0005. Images are resized to 640×640 with a batch size of 16.

We report mAP50, Precision, Recall, Params, and GFLOPs. Params and GFLOPs are measured at 640×640, and GFLOPs scale with input size, yielding 2.5 at 416×416, 5.9 at 640×640, and 15.1 at 1024×1024. The definitions of Precision, Recall, and mAP50 are given as follows:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN} \quad (11)$$

$$mAP_{50} = \frac{1}{N} \sum_{i=1}^N AP_{50}^{(i)}, \quad AP_{50} = \int_0^1 p(r) dr \quad (12)$$

B. Ablation Experiments

TABLE II
RESULTS OF ABLATION EXPERIMENTS ON THE HRSID DATASET

Models	MB-EGFEM	WaveletPool	ACF-FPN	LD-DEH	Params(M)	FLOPs(G)	mAP50%	P%	R%
A					2.58	6.3	91.2	90.9	82.5
B	✓				2.53	6.3	91.7	91.2	82.5
C		✓			2.23	5.5	91.3	93.9	80.7
D	✓	✓			2.18	5.5	92.5	89.7	85.3
E			✓		2.51	7.3	92.4	91.4	84.3
F				✓	2.26	6.0	92.4	92.9	84.0
G	✓	✓	✓	✓	1.76	5.9	93.3	92.0	86.8

Table II shows that MB-EGFEM and WaveletPool yield limited gains when used individually, improving mAP50 to 91.7% and 91.3%, respectively. In contrast, their joint integration (forming the EEWB backbone) brings a clear improvement, achieving 92.5% mAP50 and outperforming either module alone. Fig. 4 visualizes intermediate features for the baseline, MB-EGFEM only, WaveletPool only, and EEWB: MB-EGFEM enhances boundaries but introduces clutter, while WaveletPool suppresses clutter at the cost of weakened targets; EEWB produces more compact activations and clearer target-background separation, confirming their complementarity.

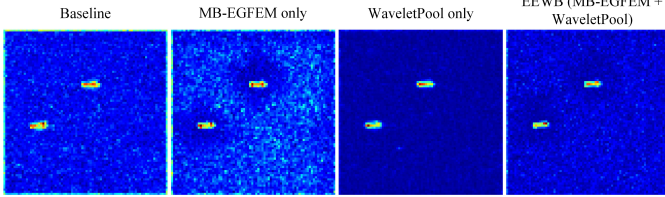


Fig. 4. All feature maps are visualized at the output of the first WaveletPool following MB-EGFEM using L2-energy maps.

ACF-FPN and LD-DEH each improve the baseline to 92.4% mAP50, achieving 84.3% and 84.0% recall, respectively. With all modules enabled, MBE-Net reaches 93.3% mAP50 and 86.8% recall while reducing complexity to 1.76M parameters and 5.9 GFLOPs. These results confirm the synergistic effect of the proposed modules in improving accuracy and efficiency.

TABLE III

SCALE-AWARE COMPARISON ON HRSID (AP@[0.50:0.95]).

Models	APS%	APM%	APL%
Baseline	49.3	81.1	76.2
MBE-Net	53.9	82.6	78.6

To further assess the multi-scale detection capability of

MBE-Net, a COCO-style scale-aware evaluation is conducted. As shown in Table III, MBE-Net achieves a clear improvement in APS over the YOLO11n baseline, without degrading performance on medium- and large-scale ships.

C. Comparison Experiments

TABLE IV
THE COMPARATIVE EXPERIMENT ON THE HRSID DATASET.

Models	Params(M)	FLOPs(G)	mAP50%	P%	R%
YOLO11n	2.58	6.3	91.2	90.9	82.5
YOLO11s	9.41	21.3	93.0	90.7	85.9
SMEP-DETR	21.11	60.7	93.2	92.7	86.5
ESarDet	6.2	23.5	93.2	-	-
LDSS-Net	2.0	8.7	91.7	89.0	91.2
MBE-Net	1.76	5.9	93.3	92.0	86.8

To evaluate MBE-Net's performance, comparative experiments were conducted on the HRSID dataset, as shown in Table IV, to assess detection accuracy and robustness in complex marine environments. MBE-Net achieves the highest mAP of 93.3% with the smallest parameter count and lowest computational cost. YOLO11n is adopted as the baseline backbone on which MBE-Net is built, while YOLO11s, a larger variant within the same model family, is included to assess the effect of model scaling. As shown in Table IV, increasing model capacity through YOLO11s substantially raises computational complexity but still fails to surpass the performance of MBE-Net, indicating that the performance gains originate from architectural design rather than model scaling. In addition, the selected baselines include representative Transformer-based detectors such as SMEP-DETR [3] as well as efficient and lightweight SAR-specific methods including ESarDet [12] and LDSS-Net [13], enabling fair comparison across different architectural paradigms and

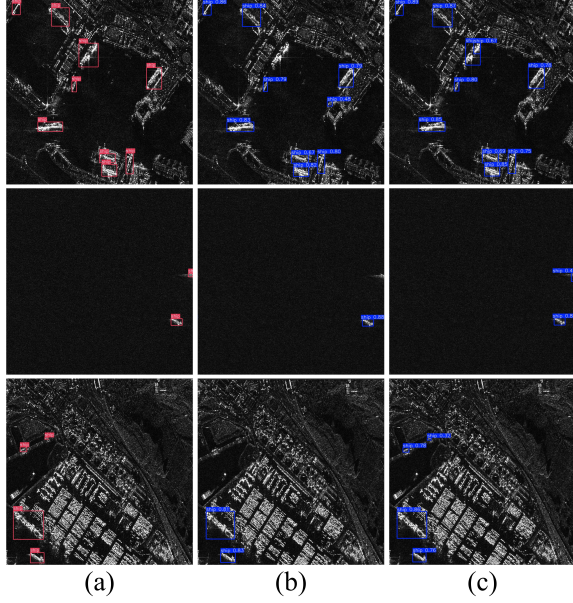


Fig. 5. Detection Performance Comparison on the HRSID Dataset.

Figure 5 compares detection results. Column (a) shows ground truth annotations, while columns (b) and (c) display outputs from YOLO11n and our method, respectively. YOLO11n struggles with false and missed detections, especially in complex scenarios with background interference or small targets. In contrast, our method improves feature extraction and robustness, yielding more accurate and stable detections.

TABLE V
THE COMPARATIVE EXPERIMENT ON THE SSDD DATASET.

Models	Params(M)	FLOPs(G)	mAP50%	P%	R%
YOLO11n	2.58	6.3	96.2	93.5	89.3
YOLO11s	9.41	21.3	97.0	93.6	92.4
FEPS-Net	37.31	-	96.0	-	-
DS-YOLO	-	24.7	96.6	95.4	91.1
LD-Det	24.4	8.1	96.8	94.8	92.4
MBE-Net	1.76	5.9	97.2	95.9	92.8

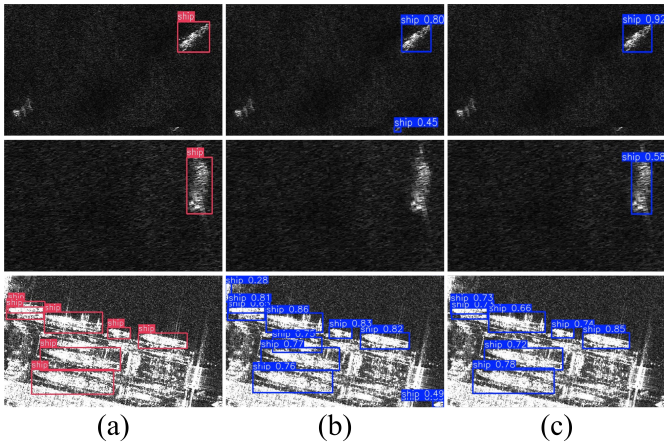


Fig. 6. Detection Performance Comparison on the SSDD Dataset.

To further validate the performance, a comparative experiment was conducted on the SSDD dataset, as shown in Table V. MBE-Net achieved a superior mAP of 97.2% with only 1.76M parameters and 5.9 GFLOPs, outperforming YOLO11n. While FEPS-Net [14] and DS-YOLO [15] also

showed comparable performance, the lightweight network LD-Det [4] achieved a mAP of 96.8% with 24.4M parameters. MBE-Net maintained a clear advantage in the accuracy-efficiency trade-off. Additionally, as illustrated in Fig. 6, MBE-Net demonstrated high-quality detection in both near-shore and off-shore scenarios, effectively reducing missed detections and false alarms.

IV. CONCLUSION

This paper presents MBE-Net, a lightweight and high-accuracy network for SAR ship detection. It addresses challenges like small targets and background clutter by integrating edge-guided and wavelet-based detail preservation, spatial refinement, and adaptive multi-scale fusion. Experiments on HRSID and SSDD show MBE-Net's superior accuracy and efficiency, making it highly promising for real-time applications. Future work will focus on further refinement and real-world deployment.

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